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Patent interface and aspects thereof

Abstract

A patient interface can have a frame supporting a sealing member. Various features of the sealing member can improve comfort and sealing performance in the context of forming seals with the nares of a user, as well as contact with other facial surfaces. The sealing member can include convex portions, concave portions and thickness variations for providing various sealing, comfort, and deformability effects.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS (1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference herein in their entireties and made a part of the present disclosure. The present application claims the priority benefit of U.S. Provisional Application No. 62/139,562, filed Mar. 27, 2015, and 62/261,177, filed Nov. 30, 2015, the entirety of each of which is hereby incorporated by reference herein.

BACKGROUND OF THE INVENTIONS

Field of the Inventions

(1) The inventions disclosed herein generally relate to interfaces for providing a supply of breathable gas to a recipient.

Description of Related Art

(2) Breathing gases can be delivered to users with a variety of different mask styles and can be delivered for a variety of different purposes. For example, users can be ventilated using non-invasive ventilation (NIV). In addition, continuous positive airway pressure (CPAP) or variable airway pressure can be delivered using masks to treat a medical disorder, such as obstructive sleep apnea (OSA), chronic obstructive pulmonary disease (COPD), or congestive heart failure (CHF).

(3) These non-invasive ventilation and pressure support therapies generally involve the placement of a user interface device, which is typically a nasal or nasal/oral mask, on the face of a user. The flow of breathing gas can be delivered from the pressure/flow generating device to the airway of the user through the mask.

(4) Typically, patient interface devices include a mask frame that supports a sealing member. The sealing member contacts the facial surfaces of the user, including regions surrounding the nose, including the nose and the nares. Because such masks are typically worn for an extended period of time, a variety of concerns must be taken into consideration. For example, in providing CPAP to treat OSA, the user normally wears the mask all night long while he or she sleeps. One concern in such a situation is that the mask should be as comfortable as possible. It is also important that the mask provide a sufficient seal against a user's face without significant discomfort.

SUMMARY OF THE INVENTIONS

(5) Accordingly, it is an object of certain embodiments of the present disclosure to provide an improved sealing member for use in a mask assembly that overcomes the shortcomings of conventional sealing members.

(6) In some configurations, a nasal seal includes a seal body formed of a soft flexible material and having a sidewall that at least partially defines an inner cavity. The seal body comprises a sealing portion having a first nostril prong and a second nostril prong. A base portion of the seal body defines a supply opening that allows a supply of breathing gases to be provided to the inner cavity. A bellows portion is defined by the sidewall. The bellows portion is located between the sealing portion and the supply opening. The bellows portion comprises an indentation that extends circumferentially around the seal body and supports both the first nostril prong and the second nostril prong.

(7) In some configurations, the indentation has a curved shape in cross-section.

(8) In some configurations, the indentation extends around an entire circumference of the seal body.

(9) In some configurations, the indentation is concave and defined between a convex base portion of the seal body and a convex sealing portion of the seal body. The convex base portion extends between the indentation and the supply opening and the convex sealing portion extending between

the indentation and the first and second nostril prongs.

(10) In some configurations, the bellows portion is configured to allow for linear compression and/or expansion of the seal body in an axial direction along an axis extending through a center of the supply opening and between the first and second nostril prongs.

(11) In some configurations, the bellows portion allows the prongs to pivot such that an angle of the prongs relative to the supply opening can vary.

(12) In some configurations, a volume of a portion of the inner cavity defined by the base portion is less than a volume of a portion of the inner cavity defined by the sealing portion.

(13) In some configurations, a curvature of the bellows portion corresponds to a curvature of a lower surface of the sealing portion adjacent the indentation, and wherein a distance between the bellows portion and the lower surface of the sealing portion is substantially constant around a circumference of the seal body.

(14) In some configurations, a thickness of the bellows portion is substantially constant throughout the bellows portion.

(15) In some configurations, the sealing portion comprises a pair thickened regions, each associated with one of the first nostril prong and the second nostril prong.

(16) In some configurations, the seal body comprises a thickened portion surrounding the supply opening.

(17) In some configurations, a middle section of the bellows portion is thinner than a section of the bellows portion adjacent the supply opening and a section of the bellows portion closest to the first and second nostril prongs.

(18) In some configurations, the supply opening is generally elliptical and a thickened portion that extends around and defines the supply opening is generally elliptical.

(19) In some configurations, a nasal seal includes a seal body formed of a soft flexible material and defines an inner cavity and a supply opening for supply of breathing gases to the inner cavity. The seal body comprises a first nostril prong, a second nostril prong and a rolling section disposed in a sidewall portion of the seal body. The rolling section being located between the supply opening and the first and second nostril prongs. The rolling section is configured to roll over on itself to reduce a distance between the supply opening and the first and second nostril prongs.

(20) In some configurations, the rolling section further comprises a rib that extends in a circumferential direction of the seal body.

(21) In some configurations, the rib is configured to contact an interior surface of the seal body to limit the roll of the rolling section and maintain the first and second nostril prongs in an operational position.

(22) In some configurations, the rib extends only partially around a circumference of the seal body.

(23) In some configurations, the rib defines a gap in the front of the seal body.

(24) In some configurations, the gap allows the rolling section in the front of the seal body to roll further towards or over the supply opening than the back of the seal.

(25) In some configurations, the rib is disposed between the supply opening and the first and second nostril prongs and projects inwardly or outwardly from the sidewall portion.

(26) In some configurations, the rib is positioned halfway between sealing surfaces of the first and second nostril prongs and the supply opening.

(27) In some configurations, a thickness of the rolling section gradually varies between the rib and the supply opening, the rolling portion being thicker adjacent the rib and thinner adjacent the supply opening.

(28) In some configurations, the rib is cantilevered from the sidewall portion, the rib tapering in thickness between a thicker portion adjacent the sidewall portion and a thinner portion further from the sidewall portion.

(29) In some configurations, the supply opening is kidney bean shaped and generally follows the overall shape of the seal body.

- (30) In some configurations, an intersection between the rolling section and an end of the seal body that defines the supply opening is curved in a lateral direction of the nasal seal.
- (31) In some configurations, a nasal seal includes a seal body formed of a soft flexible material and defining an inner cavity that receives a supply of breathing gases. The seal body comprises a first nostril prong and a second nostril prong. Each of the first and second nostril prongs are non-conical and asymmetrical.
- (32) In some configurations, the seal body defines a supply opening through which the supply of breathing gases passes into the inner cavity, wherein the supply opening is non-circular in shape.
- (33) In some configurations, the supply opening is bean-shaped.
- (34) In some configurations, each of the first and second nostril prongs comprises a medial portion, wherein the medial portion is oriented at an angle relative to a central plane of the seal body.
- (35) In some configurations, the medial portion comprises an upper portion and a lower portion, wherein the angle relative to the central plane of the lower portion is greater than the angle relative to the central plane of the upper portion.
- (36) In some configurations, a rolling portion is defined by a sidewall portion of the seal body and configured to roll over on itself, the rolling portion comprising an upper portion configured to be distal of an upper lip of a user and a lower portion configured to be proximate the upper lip of the user. The upper portion provides less resistance to roll than the lower portion.
- (37) In some configurations, a proximal wall of the seal body comprises a concave portion extending in a lateral direction of the seal body and having a shape that generally conforms to a shape of an upper lip of a user.
- (38) In some configurations, the first nostril prong comprises a first sealing surface and the second nostril prong comprises a second sealing surface, wherein portions of the first and second sealing surfaces are convex.
- (39) In some configurations, the seal body comprises a central wall portion extending between and connecting the first nostril prong and the second nostril prong, wherein the central wall portion is spaced deeply from an end surface of the first and second nostril prongs such that contact between the central wall portion and a septum of the user is reduced or eliminated.
- (40) Various features, aspects and advantages of the present embodiments can be implemented in any of a variety of manners. For example, while several embodiments will be described herein, sets or subsets of features from any of the embodiments can be used with sets or subsets of features from any of the other embodiments.
- (41) The term “comprising” is used in the specification and claims, means “consisting at least in part of” When interpreting a statement in this specification and claims that includes “comprising”, features other than that or those prefaced by the term may also be present. Related terms such as “comprise” and “comprises” are to be interpreted in the same manner.
- (42) In this specification where reference has been made to patent specifications, other external documents, or other sources of information, this is generally for the purpose of providing a context for discussing the features of the inventions. Unless specifically stated otherwise, reference to such external documents is not to be construed as an admission that such documents, or such sources of information, in any jurisdiction, are prior art, or form part of the common general knowledge in the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other features, aspects and advantages of the present inventions are described below with reference to the following drawings.
- (2) FIG. 1 is a front, top, and right side perspective view of a patient interface including a seal, a

conduit connector, and a gas supply conduit, configured in accordance with an embodiment.

(3) FIG. 2 is a side elevational view of the interface of FIG. 1.

(4) FIG. 3 is a view of the interface as viewed along line 3, of FIG. 2.

(5) FIG. 4 is a front, left, and top perspective view of the interface of FIG. 1.

(6) FIG. 5 is an embodiment of a seal body that can be used with the interface of FIGS. 1-4, viewed in a direction such as that on by arrow 3, in FIG. 2.

(7) FIG. 6 is an enlarged schematic and partial sectional view of nostrils of a human patient.

(8) FIG. 7 is a sectional view of the seal of FIG. 5 engaged with human nostrils schematically illustrated in FIG. 6.

(9) FIG. 7a is a sectional view of the embodiment of FIG. 7, with certain lines removed.

(10) FIG. 7b this is a schematically diagram illustrating variations in dimensions of nares amongst some human populations.

(11) FIG. 7c is a view of the seal of FIG. 5, which schematic indications of examples of optional locations of sealing between the seal and the nares of differently-shaped human noses.

(12) FIG. 7d is a top plan view of the seal with the schematic indications of FIG. 7c.

(13) FIG. 7e is a left side elevational view of the seal with the schematic indications of FIG. 7c.

(14) FIG. 7f is a top, front, and left side perspective view of a modification of the seal of FIG. 5.

(15) FIG. 7g is a sectional view of the seal of FIG. 7f, taken along section line 7g.-7g.

(16) FIG. 7h is a sectional view of the seal of FIG. 7f, taken along section line 7h.-7h.

(17) FIG. 8 is a perspective and partial sectional view of the seal of FIG. 5.

(18) FIG. 9 is another partial sectional and perspective view of the seal of FIG. 5.

(19) FIG. 10 is a perspective view of the seal of FIG. 5.

(20) FIG. 11 is a front elevational view of the seal of FIG. 5.

(21) FIG. 12 is a perspective view of a modification of the seal of FIGS. 5-11.

(22) FIG. 13 is a side elevational view of the seal of FIG. 12.

(23) FIG. 14 is a sectional view of the seal taken along line 14.-14. of FIG. 13.

(24) FIG. 15 is an enlarged view of a portion of the sectional view of FIG. 14.

(25) FIG. 16 is a sectional view of the seal taken along line 16.-16. of FIG. 13.

(26) FIG. 17 is a top plan view of seal of FIG. 12.

(27) FIG. 18 is a skewed view of the seal of FIG. 12 as viewed in the direction of arrow 3, of FIG. 2.

(28) FIG. 19 is a sectional view of the seal taken along line 16-16 of FIG. 13 and illustrating a lateral movement.

(29) FIG. 20 is a top, front and left side perspective view of the seal member of FIG. 12, including shaded contours corresponding to optional thicknesses.

(30) FIG. 21 is a left side elevational view of the seal with the thickness shading of FIG. 20.

(31) FIG. 22 is a bottom, rear and left side perspective view of the seal with the thickness shading of FIG. 20.

(32) FIG. 23 is a rear elevational view of the seal and with the thickness shading of FIG. 23.

(33) FIG. 24 is a skewed view of the seal of FIG. 12 taken along a direction identified by arrow 3, of FIG. 2, and including schematic lines and shading as optional boundaries of different portions of the seal.

(34) FIG. 25 is a top plan view of the seal of FIG. 20.

(35) FIG. 26 is a side elevational view of the seal of FIG. 20.

(36) FIG. 27 is a bottom plan view of the seal of FIG. 20.

(37) FIG. 28 is a front elevational view of the seal of FIG. 20.

(38) FIG. 29 is perspective view of the seal of FIG. 20.

(39) FIG. 30 is a rear, right, and bottom perspective view of the seal of FIG. 20.

(40) FIG. 31 is a perspective view of the seal of FIG. 20.

(41) FIG. 32 is a perspective view of an example embodiment of a seal including a bellows.

- (42) FIG. 33 is a sectional view of the seal of FIG. 32 showing a longitudinal axis and possible pivoting movement of the seal.
- (43) FIG. 34 is a sectional view of the seal of FIG. 32 showing an outline of a compressed profile of the seal.
- (44) FIG. 35 is a force vs. compression graph.
- (45) FIG. 36 is a sectional view of the seal of FIG. 32.
- (46) FIG. 37 is another sectional view of the seal of FIG. 32.
- (47) FIG. 38 is a sectional view of the seal of FIG. 32 showing widths of various parts of the seal.
- (48) FIG. 39 is a sectional view of the seal of FIG. 32 showing an outline of a splayed profile of the seal.
- (49) FIG. 40 is sectional view of the seal of FIG. 32 showing an outline of an inflated profile of the seal.
- (50) FIG. 41 is a perspective view of an embodiment of the seal of FIG. 32 including a thickened region.
- (51) FIG. 42A is a sectional view of the seal of FIG. 32.
- (52) FIG. 42B is a view of the seal of FIG. 32 as viewed along line 3, of FIG. 2.
- (53) FIG. 42C is a side elevational view of the seal of FIG. 32.
- (54) FIGS. 42D, 42E and 42F show various views of the seal of FIG. 32 showing interior volumes of various regions of the seal.
- (55) FIGS. 42G, 42H and 42I show various views of the seal of FIG. 32 showing interior volumes of various regions of the seal.
- (56) FIG. 43 is a sectional view of an example embodiment of a seal having a rolling section.
- (57) FIG. 44 is a sectional view of the seal of FIG. 43 showing an outline of the rolling section of the seal in a rolled state.
- (58) FIG. 45 is a detail view of a portion of a sectional view of the seal of FIG. 43 showing an internal rib.
- (59) FIG. 46 is a detail view of a portion of a sectional view of the seal of FIG. 43 showing varying thickness in the rolling section.
- (60) FIG. 47 is a theoretical force v. displacement profile of the seal of FIG. 43.
- (61) FIG. 48 shows a force profile of the seal of FIG. 43.
- (62) FIG. 49 shows a heat map illustrating the thickness of various regions of the seal of FIG. 43.
- (63) FIGS. 50A, 50B and 50C illustrate front elevational views of the seal of FIG. 43.
- (64) FIG. 51 is a sectional view of the seal of FIG. 43 showing a depth of the rolling section.
- (65) FIG. 52 is a sectional view of the seal of FIG. 43 showing an outline of the seal in a splayed state.
- (66) FIG. 53A is a top view of the seal of FIG. 43.
- (67) FIG. 53B is a detail view of a connector receiving portion of the seal of FIG. 43.
- (68) FIGS. 54A, 54B and 54C are various views of the seal of FIG. 43.

DETAILED DESCRIPTION

(69) FIGS. 1-4 schematically illustrate a patient interface **100** that can include one or any combination of certain features, aspects, and advantages described herein. The interface **100** can be used to supply pressurized breathing gases to a recipient, such as a user or a patient. The interface **100** can be used for providing breathing gases to a recipient in situations where significant pressure changes may be encountered. For example but without limitation, the interface **100** can be used for delivering continuous positive airway pressure (CPAP).

(70) With continued reference to FIGS. 1-4, the patient interface **100** is shown separated from a patient who might wear the interface **100**. Some aspects of the patient interface **100**, and variations on each aspect, have been described in U.S. patent application Ser. No. 12/945,141, filed Nov. 12, 2010, which is hereby incorporated by reference in its entirety. The interface **100** can be in the form of a mask **102**. In some configurations, a strap **104** can be used to secure the mask to a patient. In

some configurations, the interface can also comprise a flexible supply conduit **106** which can connect to the mask **102**.

(71) The mask **102** can be configured to fit over or overlies both nostrils of a patient. The strap **104** can be wrapped around the user's head in a simple loop above the user's ears, although other configurations and uses for the strap **104** are also available. In some configurations, the mask **102** can include lateral extending portions (not shown) configured to curve around toward each lateral side of a nose of a patient. Such lateral portions can form a perimeter seal on outwardly facing surfaces of flanks of the nose. On the other hand, the embodiments disclosed herein are described in the context of masks, which can be referred to as nasal pillow masks, which do not include laterally extending portions configured to curve around toward each lateral side of the nose of a patient to contact or form a perimeter seal on outwardly facing surfaces of flanks of the nose of a patient, such as the wings **130** disclosed in PCT Application No. PCT/NZ2014/000150 filed Jul. 17, 2014. Some of the embodiments disclosed herein can provide beneficial advantages of enhanced seals sealing performance and comfort while eliminating such laterally extending portions.

(72) The flexible conduit **106** can depend from a central connection **112**. In some configurations, the central connection **112** can be positioned at a frontal portion of the mask **102**. The central connection **112** can include a swiveling elbow. Such a swiveling elbow can be configured to allow the flexible conduit **106** to pivot relative to the mask **102**. By enabling pivoting, the elbow can help the interface **100** to better adapt to the sleeping positions of a patient. In some configurations, the central connection **112** may comprise a ball joint so that the elbow can pivot about axes parallel to and perpendicular to its connection with the mask **102**. In some configurations, the central connection **112** may comprise a swivel or swivel elbow.

(73) The illustrated mask **102** can generally comprise a seal **114** and a body or frame **116**. The seal **114** and the frame **116** can be connected in any suitable manner.

(74) The seal **114** can define a supple pocket or envelope that can contain a recess region. In some configurations, the seal **114** can comprise a low wall thickness and can be formed of any suitable material. For example but without limitation, the seal **114** can be formed of latex, vinyl, silicone or polyurethane. In some configurations the wall thickness can be below about 0.5 millimeters and could be lower than about 0.2 millimeters in some regions and in some configurations.

Additionally, as described below, some portions of the seal **114** can include thickened regions having thicknesses of 0.7 millimeters, 0.9 millimeters, 1.2 millimeters, as well as greater thicknesses. In some configurations, the seal **114** can be formed of a material having sufficient elasticity and yield strength so that the combination renders the seal **114** supple. The seal **114** can be configured to withstand repeated drastic deformations without failure.

(75) With continued reference to FIGS. 1-4, the seal **114** can include one or two nostril prongs **120**, **121**. The nostril locators **120**, **121**, can protrude from the seal **114**. In some configurations, the nostril locators **120**, **121** extend generally upwardly and rearwardly from a proximal wall **124** of the seal **114**. As used herein, the term “proximal” is intended to refer to portions of the mask **102** that are proximate or closer to a patient during use. On the other hand, the term “distal” is used to refer to portions of the mask **102** which are disposed further away from a patient during use.

(76) In the configurations illustrated in FIGS. 1-4, the nostril prongs **120**, **121** are formed integrally (i.e., in a single monolithic piece) with the seal **114**. Each nostril prong **120**, **121** can comprise an outward aperture **118**, **119** through which gas can be supplied from the flexible conduit **106**, to nostrils of the patient. In some configurations, the gas can be supplied from within the pocket or envelope defined by the seal **114**. In other configurations, the gas that is supplied can be separate from the gas supplied to the pocket or envelope defined by the seal **114**.

(77) The seal **114** can also be considered as having a distal wall **122**. As such, an outer surface of the distal wall **122** would face away from the user during use, while an outer surface of the proximal wall **124** would face the user.

(78) The seal **114** can also include left and right lateral walls **125**, **126** disposed on the left and right

sides, respectively, of the nostril prongs **120**, **121**. Additionally, the seal **114** can include an intermediate wall portion **127** extending between the nostril prongs **120**, **121**. The walls **122**, **124**, **125**, **126** extend from the nostril prongs **120**, **121** toward the flexible conduit **106**, along the direction of arrow A in FIG. 2. Thus, proximal portions of the walls **122**, **124**, **125**, **126** can be considered as connecting with the nostril prongs **120**, **121**. Distal ends of the walls **122**, **124**, **125**, **126** can extend to an inlet orifice **130** (FIG. 7). The inlet orifice **130** can be connected to the frame **116** and/or directly to the flexible conduit **106**.

(79) As such, the envelope or pocket described above can be defined within at least the distal wall **122** and the proximal wall **124** and/or optionally, the lateral walls **125**, **126**, or a combination thereof. The seal **114** can be configured to achieve a desired seal without wrapping around the tip or lower portion (e.g., locations below the bridge) onto substantial portions of the outer surfaces a user's nose. However, in some embodiments, the seal **114** can be designed to wrap around the tip or lower portion of a user's nose. The seal **114** can also be considered as including a central portion **128** which can be considered as including the nostril prongs **120**, **121**, and the central wall **127**. The central portion **128** can, in some embodiments, include sealing surfaces of the mask **102** which form seals with the nostrils of a user during use. The central wall **127** can be disposed significantly deeper and further distally from the orifices **118**, **119** than the other portions, such as the sealing surface within the central portion **128**. As such, the central wall **127** provides improved clearance for the user's septum, reducing contact and potential discomfort or irritation.

(80) As described above, optionally, a substantial portion of the seal **114** can be supple. For example, a region surrounding the nostril locators, such as the central region **128**, can be more supple than distal portions of the walls **122**, **124**, **125**, **126**. Further, optionally, the proximal wall **124** and the nostril prongs **120**, **121** can be very supple so that they can expand to conform to the contours of a user's nares. The supple portions of the seal **114** can be of sufficient dimension and shape that when positive pressure of gases within the seal **114** can partially expand portions of the seal **114** to enhance a seal between the seal **114** and the user's face, for example, nostrils of the user. Additionally, portions of the seal, for example, along proximal portions of the proximal wall **124** can rest against or provide the sealing engagement with a portion of the user's upper lip.

(81) Optionally, portions of the seal **114** can have an increased rigidity to improve form, fit, and function of the mask **102**. For example, portions of the seal **114** can be made significantly stiffer to provide control of ballooning of other regions of the seal **114**. Further, distal portions of the walls **122**, **124**, **125**, **126**, for example, regions or areas adjacent to or proximate to the inlet orifice **130** (FIG. 7) can be less supple. Thus, for example, one or more the walls **122**, **124**, **125**, **126** can have a decreasing suppleness from the proximal portions thereof to the distal portions thereof. Such less supple regions can be formed of a different material or can be formed of the same material but with an increased thickness.

(82) The frame **116** can have any arrangement for fixation or connection to the seal **114**. In some configurations, an annular wall can extend from a proximal side of the frame **116** around a perimeter of an opening extending to the connector **112**. The annular wall can include an outwardly extending lip. The inlet orifice **130** of the seal **114** can engage over the outwardly extending lip of the annular wall. In some configurations, the inlet orifice **130** of the seal **114** can be stretched to fit over the annular wall. The inlet orifice **130** of the seal **114** can be provided with a thickened or reinforced wall section, for example but with limitation. In some configurations, an extended portion of the inlet orifice **130** can be rolled up over the annular wall of the frame **116**. In other configurations, the seal **114** can be provided with a portion of a connector on at least one of the frame **116** and the conduit **106** can include a complimentary connector portion.

(83) The frame **116** can be configured to be minimal in size. For example, a small sized frame **116** can enable a clear field of vision for the user and can allow a user to wear glasses while wearing the interface **100**. The frame **116** can be formed of an elastomeric material, which can allow the frame **116** to flex to conform slightly to the face of the user. The frame **116**, however, can provide some

support for the seal **114**. By providing support for the seal **114**, the seal **114** can be more effectively pressed into contact with the face and nose of the user. The frame **116** can be formed by injection molding, preferably from an elastic material, such as silicone or polyurethane, for example but without limitation. In some configurations, the frame **116** can be formed of more rigid materials, such as polycarbonate, polyester polystyrene, or nylon, for example but without limitation.

(84) In use, portions of the seal **114** can be inflated and thus expanded outwardly. Pressure within the seal **114** (e.g., inflated from the flow of pressurized gases supplied to the patient interface **100**) can press the sealing surfaces of the seal **114** against the skin of a user and to further conform to contours of the outside surfaces of the nose of the user, to surfaces of the lower portion of the nose of the user, to various portions of the nares of the user, as well as surfaces of the upper lip of the user immediately below the nose.

(85) With reference to FIG. **6**, the lower portion of a human nose **200** (illustrated in front elevation and partial cross section) can be considered as including the walls and surfaces and portions described below, the names of which are included herein for facilitating understanding of the present disclosure.

(86) With continued reference to FIG. **6**, the lower end of a human nose **200** can include a right nostril lateral wall **202**, a left nostril lateral wall **204**, and a septum **206**. The right nostril lateral wall **202** extends from a forward facing surface of the patient's face, to the tip of the nose. Similarly, the left nostril lateral wall **204** extends from a left side of the user's face toward the tip of the nose. The septum **206** extends from the user's face to the tip of the nose, and between the left and right nostril lateral walls **202**, **204**. As such, the right nostril lateral wall **202** and the septum **206** define the right nostril **208** of the user. Similarly, the left nostril lateral wall **204** and the septum **206** define the left nostril **210** of the user.

(87) The lower ends of the lateral walls **202**, **204** and the septum **206** can be also considered as forming the nares of the user, including outer, inner and lower surfaces thereof.

(88) For example, the lower portion of the right nostril lateral wall **202** can be considered as having lateral outer surface **212**, bottom surface **214**, and intermedial surface **216**. Similarly, the lower portion of the left nostril lateral wall **204** can be considered as including outer lateral surface **218**, bottom surface **220**, and intermedial surface **222**.

(89) The septum **206** can be considered as including right lateral surface **224**, bottom surface **226** and left lateral surface **228**.

(90) With reference to FIG. **5**, the right and left nostril prongs **120**, **121** can also be considered as including lateral, medial, proximal, and distal portions. For example, the right nostril prong **120** can be considered as including a proximal portion **240**, a lateral portion **242**, a distal portion **244** and a medial portion **246**. Similarly, the left nostril prong **121** can be considered as including a proximal portion **250**, a lateral portion **252**, a distal portion **254** and a medial portion **256**.

(91) With continued reference to FIG. **7**, as noted above, the seal **114** can be configured to be sufficiently supple that it can be inflated and thus expanded by positive pressure within the seal **114**. Thus, for example, as illustrated in FIG. **7**, such pressure can beneficially push the lateral portions **242**, **252** of the right and left nostril prongs **120**, **121** into enhanced contact with the medial surfaces **216**, **222**, and optionally further contact with the bottom surfaces **214**, **220**, and further optionally, the lateral surfaces **212**, **218**, optionally without making contact with substantial portions of the lateral surfaces **212**, **218**. Similarly, the medial portions **246**, **256** can be pushed inwardly by the positive pressure, thereby increasing contact and sealing effect against the lateral walls **224**, **228**.

(92) With continued reference to FIGS. **7** and **7a**, as described above, the nostril prongs **120**, **121** can be configured to seal against the septum **206** and the lateral nostril walls **202**, **204**, differently. For example, as described above, the medial portions **246**, **256** of the nostril prongs **120**, **121** can be configured, oriented, and/or shaped to press more against the lateral surfaces **220a**, **224** of the septum **206** and reduce or minimize contact with bottom surface **226** of the septum **206**. Optionally,

the central wall **127**, as noted above, can be spaced deeply between the nostril prongs **120**, **121**, so as to reduced or eliminate the likelihood of contact with any or a substantial portion of the bottom surface **226** of the septum **206**. The degree to which the medial portions **246**, **256** press against the lateral surfaces **224**, **228** and less on the bottom surface **226** can be contrasted with the configuration, shape, and orientation of the laterals portions **242** and **252**.

(93) Optionally, the lateral portions **242**, **252** can be shaped, configured and/or oriented so as to contact less of the medial portions **216**, **222** of the right and left nostril walls **202**, **204** and more of the bottom surfaces **214**, **220** and the lateral surfaces **212**, **218**, during use, as compared with the contact of the medial portions **246**, **256** with the corresponding portions of the septum **206**, described above.

(94) Asymmetric Prongs

(95) With reference to FIGS. 7 and **7a**, optionally, in some embodiments, the nostril prongs **120**, **121** can be configured to provide a differential sealing contact with a user's nose **200**, with an asymmetric prong shape. Certain aspects of the asymmetry of the prongs **120**, **121** can be described with reference to a central plane P bisecting the seal **114** in a direction directly between the prongs **120**, **121**. For example, as shown in FIG. **7a**, the medial portion **256** of the nostril prong **121** extends in a direction that is nearly vertical or in other words, parallel, nearly parallel, substantially parallel, or within approximately 10 degrees of the central plane P. In some embodiments, the medial portion **256** of nostril prong **121** extends at an angle, with respect to central plane P, of approximately 10 degrees, 0-15 degrees, 0-5 degrees, 5-10 degrees, 10-15 degrees, 5-15 degrees, greater than 0 degrees, or various sub-ranges of 0-15 degrees. The slope of the medial portion **256** is represented by line **270**. With regard to this description of the seal **114**, and other descriptions set forth below, in some instances, features are only described in connection with one of the prongs **120**, **121**, but it is to be understood that both of the prongs **120**, **121** can have the same shape, and thus the descriptions would apply equally to both prongs **120**, **121**. With continued reference to FIG. **7a**, in some embodiments, the medial portion **256** comprises an upper portion **256a** and a lower portion **256b**. The upper portion **256a** can be angled, with respect to plane P, at approximately 10 degrees, 0-15 degrees, 0-5 degrees, 5-10 degrees, 10-15 degrees, 5-15 degrees, greater than 0 degrees, or various sub-ranges of 0-15 degrees. The lower portion **256b** of the medial portion **256** can be angled, with respect to plane P, at approximately 20 degrees, 0-30 degrees, 0-20 degrees, 10-30 degrees, 15-25 degrees. In some embodiments, the upper portion **256a** of the medial portion **256** comprises up to one half of the medial portion **256**.

(96) In contrast to the slope of the medial portion **256**, the slope of the lateral portion **252** is not nearly parallel, approximately parallel or substantially parallel with the central plane P. Rather, the slope of the lateral portion **252** is transverse to the plane P and optionally can be nearly perpendicular to the plane P. In FIG. **7a**, the slope of the lateral portion **252** is identified by the line **272**. The line **272** extends at an angle of about 75 degrees relative to the plane P. In some embodiments, the slope **272** can be within the range of approximately 60-90 degrees relative to the plane P. In some embodiments, the angle is approximately 70 degrees, 60-70 degrees, 70-80 degrees, 80-90 degrees, 60-75 degrees, 75-90 degrees, less than 90 degrees, or various sub-ranges of 60-90 degrees.

(97) With continue reference to FIGS. 5 and **7a**, the difference in shape between the medial portions **246**, **256** and the lateral portions **242**, **252** also extends, optionally, partly into the distal and proximal portions **244**, **254**, **240**, **250**. Additionally, the distal and proximal portions **244**, **254** can extend through contours so as to smoothly transition between the different shapes of the medial portions **246**, **256** and the lateral portions **242**, **252**. For example, portions of the proximal and distal portions **250**, **254** adjacent to the lateral portion **252** can have a similar shape, while portions of the proximal and distal portions **250**, **254** adjacent to the medial portion **256** can have a shape similar to the medial portion **256**, so as to provide a smooth transition around the periphery of the prongs **120**, **121**.

(98) Convex Sealing Surface

(99) Optionally, the seal **114** can include further features for sealing differently against the left and right lateral walls **202**, **204** compared to sealing against the septum **206**. For example, with reference to FIG. **7a**, as noted above, the medial portions **246**, **256** of the prongs **120**, **121** extend generally vertical or parallel with the central plane P. In contrast, the lateral portions **242**, **252** can include a convex sealing surface positioned for contact with the lower portion of the patient's lateral nostril walls **202**, **204**.

(100) As shown in FIG. **7a**, a proximal end of the lateral portion **242** includes a concave portion **280**. Distal from the portion **280** is a convex portion **282**. The concave and convex portions **280**, **282** fall within the central area **128** which can be considered as comprising the sealing surfaces of the seal **114**, described above with reference to FIG. **3**.

(101) In some embodiments, the concave portion **280** can be smaller than the convex portion **282**. Thus, the concave portion may only act as a sealing surface on patients with particularly small nares. The convex portion **282** can be larger to accommodate a larger range of different size nares, and preferentially sealing against such nares with the convex portion **282** which is convex in a relaxed state, but may be deformed into a concave state upon application against a patient's nose **280**. In this respect, in at least some patients, the concave portion **280** may be a locating or alignment feature and the convex portion **282** is a sealing surface. Extending in a substantially horizontal angle can help limit the depth to which the nostril prongs **120**, **121** extend into the patient's nostrils **208**, **210**. Inserting prongs too deeply into a patient's nose can cause contact with sensitive internal surfaces of the nares and cause irritation and discomfort for the patient. Thus, the depth of the prongs **120**, **121** and the extent to which prongs **120**, **121** would extend into a nose can be balanced with the need for stability provided by the prongs **120**, **121**, which is enhanced when the prongs **120**, **121** sit far enough within the nostrils that they aren't too easily displaced when the mask **102** moves on the patient's face. In a non-limiting example, the distance between the medial surface **246** and the lateral end of the concave portion **280** is approximately 12.7 millimeters, and the width of the convex portion **282** can be approximately 6.8 millimeters, as measured in the directions identified in FIG. **7a**.

(102) As noted above, the prongs **120**, **121** and have a concave sealing area which can be configured to provide reduced contact area with nares a user, so as to reduce irritation of the nares. Measurements of human noses have revealed that among at least some groups of human populations, the width x of a human septum **226**, is fairly constant, i.e., varies little. On the other hand, such measurements have revealed that the length y of human nares, as well as the outermost width of human nares, vary relatively more. Thus, the shape of the openings and the prongs **120**, **121** themselves, are generally non-conical and asymmetric so as to better accommodate these variations. For example, as noted above, because the width x of the human septum **226** varies little, the shape of the prongs **120**, **121**, in those areas which would contact a septum **226**, such as the medial walls **256**, **246**, can be fairly vertical so as to reduce the likelihood of irritation caused on the nares, such as the lateral wall portions **224**, **228**, and the lower wall of the septum **226**.

(103) For example, with reference to FIG. **7c**, longer and narrower nostrils can tend to seal around the prong **120** along sealing line **286**, smaller nostrils can tend to seal around the prong **120** along sealing line **287**, and wider nostrils can tend to seal around the prong **120** along sealing line **288**. With regard to FIG. **7d**, comparison of the lines **286**, **287**, **288**, shows a distal-proximal variation Vz (generally corresponding to the "z" dimension identified above with reference to FIG. **7b**) in the lateral locations of the lines **286**, **287**, **288**, on the lateral portions **242**, **252**, of the prongs **120**, **121**.

(104) Similarly, with reference to FIG. **7e**, comparison of the lines **286**, **287**, **288** shows a medial-lateral variation VY (generally corresponding to the "y" dimension identified above with reference to FIG. **7b**) of the locations of the feeling lines **286**, **287**, **288**, on the distal portions **244**, **254**, of the prongs **120**, **121**.

(105) FIGS. **7f-7h** illustrate a modification of the seal **114**, identified generally by the reference

numeral **114a** that can also include the features described above with reference to seal **114** as well as the features described below with reference to seal **1114**.

(106) Semi-Inflating Seal

(107) As noted above, the sealing surfaces extending around the nostril prongs **120**, **121** can be made sufficiently flexible and supple so as to inflate to a degree when subject to positive air pressure within the seal **114**. Such inflation allows for more positive engagement between the sealing surfaces and the lower surfaces of the patient's nose. More specifically, the proximal, lateral, distal, and medial portions **240**, **242**, **244**, **246** of the nostril prong **120**, which comprise the sealing surfaces within the area **128**, can be made sufficiently thin and/or supple so as to partially inflate and deflect under positive pressure generated within the seal **114**. As such, these portions **240**, **242**, **244**, **246** can deflect to a degree, i.e., outwardly, so as to provide better positive sealing and engagement with the medial surface **228** of the septum **206** and the medial, bottom, and lateral surfaces **222**, **220**, **218** of the patient's lateral nostril wall **204**.

(108) Lip Engagement

(109) Optionally, the seal **114** can also include enhanced comfort and/or engagement with an upper lip of a patient. For example, with reference to FIG. 5, the proximal wall **124** can include a concave portion **290** extending in a lateral direction, transverse to the central plane P. This concave portion **290** can have a shape that generally conforms to a typical shape of a human's upper lip, and particularly the portion of the upper lip proximal to the lower end of the user's nose. Typically, the upper lip of a human extends in a lateral direction and has a generally convex shape. Thus, the concave portion **290** is configured to fit a typical human upper lip configuration. In some embodiments, the concave portion **290** extends over a distance **292** which is approximately equal to a distance between the centers of the nostril prongs **120**, **121**. However, other lengths of the concave portion can also be used.

(110) In some embodiments, the concave portion **290** includes a radius of curvature **291** of approximately 12.5 mm, 10-15 mm, 12-13 mm, or various sub-ranges thereof. However, other configurations can also be used. Optionally, the thickness of the proximal wall **124**, in the area of the concave portion **290**, can be reduced so that it conforms more readily to the lip geometry of individual users. In a non-limiting exemplary embodiment, the thickness of the proximal wall **124** is approximately 0.7 mm.

(111) Rolling Section

(112) Optionally, the seal **114** can include a configuration for a preferential rolling deformation in the vicinity of the connection between the side walls of the seal **114** and the inlet orifice **130**. For example, as described above with reference to FIG. 2, the proximal, lateral, and distal walls **124**, **125**, **126**, **122**, extend from the nostril prongs **120**, **121** to the inlet orifice **130**. The distal ends of the walls **124**, **125**, **126**, **122**, include rolling portions formed in the vicinity of the inlet orifice **130**, for example, including the last few millimeters of the walls. For example, the walls can extend toward the inlet orifice **130** along the direction A (FIGS. 2 and 7), overshoot the connection point to the inlet orifice **130**, and curl back, in a direction opposite to the direction of arrow A. This overshoot and curl configuration generates an area in these walls which tends to deform first when forces are applied, causing a rolling deformation effect.

(113) With reference to FIG. 11, the rolling portions at the distal ends of the walls **122**, **124**, **125**, **126** can be considered as being within the area **300**. Thus, the walls **122**, **124**, **125**, **126** can have associated rolling portions **302**, **304**, **306**, **308**, within the rolling area **300**, as illustrated in FIG. 11. The cross-section of FIG. 7 illustrates a configuration of the portions **306**, **308**.

(114) As shown in FIG. 7, the wall **126** extends distally from the nostril prong **120** to the rolling portion **306**. The rolling portion **306** extends beyond the connection point **310**; i.e., lower than the connection point **310** as viewed in FIG. 7, then extends upwardly in a direction opposite to the direction of arrow A, to the connection point **310**. This overshoot and curled back configuration of the rolling portion **306** can provide for enhanced and preferential deformation and rolling of the

rolling area **300**.

(115) FIG. **9** illustrates rolling portions **302** and **304**. As shown in FIG. **9**, the rolling portion **302** is generally larger than the rolling portion **304**. One characteristic of the rolling portion **302** that is larger than the rolling portion **304** is the magnitude of the overshoot, identified generally by the reference numeral **320**. As illustrated in FIG. **9**, the overshoot **320** is larger than the overshoot **322**. In some embodiments, the overshoot **320** can be approximately 1 millimeters and the overshoot **322** can be approximately 0.5 millimeters.

(116) Another differentiating characteristic of the rolling portions **302** and **304** is the radial dimensions of the rolling portions **302**, **304**. For example, as shown in FIG. **9** the rolling portion **302** has a radial dimension **324** that is larger than a corresponding radial dimension **326** of the rolling portion **304**. In some embodiments, the radial dimension **324** is approximately 8 mm. In some embodiments, the radial dimension **326** is approximately 3 mm.

(117) These dimensional differences provide a differential rolling propensity of the rolling portions **302** and **304**. For example, the larger overshoot **320** and larger radial dimension **324** can cause the rolling portion **302** to roll more readily and through larger magnitudes of rolling deformation than the rolling portion **304**.

(118) As such, although the rolling area **300** can extend around the entire periphery of the inlet orifice **130**, it can be configured to more readily roll and deform at the distal portions of the seal **114**. Thinner wall sections, near the inlet orifice **130**, for example, in the rolling area **300**, can help determine points at which rolling deformation is initiated or greater in magnitude.

(119) The rolling area **300** can act as a shock absorber that can isolate effects of hose drag, for example, forces on the conduit **106** and tension that may be generated in the head gear or strap **104** (FIG. **1**), preventing those forces or tensions from affecting the seal performance. This can help reduce the likelihood that the nostril prongs **120**, **121** are pulled away from the patient's nose **200**, thereby preventing or reducing the likelihood of leaking.

(120) With reference to FIG. **9**, during use, the inlet orifice **130** can be relatively close to the upper lip of the user. Thus, lower portions of the frame **116** and connector **112** can also be close to the upper lip of the user. Thus, by providing the rolling section **304** with smaller dimensions and or larger thicknesses can provide the optional benefit of a lower propensity for rolling. Optionally, the rolling portion **304** can be configured for very little rolling in this area. Such rolling portions **300** can help reduce the likelihood that the seal **114** will collapse on the patient's lip or other parts of the face, thereby providing an enhanced degree of stability that helps maintain the position of the sealing surfaces relative to the lower nasal surfaces of the user's nose **200**. Optionally, the rolling portion **302** can have a continuous curve to thereby increase the likelihood that the portion **302** will roll rather than collapse or fold during use.

(121) FIGS. **12-31** illustrate a modification of the seal **114**, identified generally by the reference numeral **1114**. Parts, components, and features of the seal **1114** which are similar or the same as corresponding parts, components, or features of the seal **114** are identified with the same reference numeral, except that 1000 has been added thereto.

(122) With reference to FIGS. **12-19**, the seal **1114** can include the same features of the seal **114** noted above with respect to asymmetric prongs, convex sealing surfaces, semi-inflating, lip engagement, and rolling. Thus, those features which can optionally be embodied in the seal **1114**, are not further described below.

(123) The seal **1114** includes an optional configuration for enhancing response to lateral forces and a rolling movement accommodated by the lateral walls **1125**, **1126**.

(124) With reference to FIG. **16**, the lateral wall **1125** can be considered as extending from the nostril prong **120**, in the direction of arrow A, toward the inlet orifice **1130**. The seal **1114** can optionally include the rolling portion **306** in the area proximal to the inlet orifice **1130**.

Additionally, the seal **1114** can include an additional rolling portion **1400** in the lateral wall **1125**. The seal **1114** can include an additional rolling portion **1400** in the lateral wall **1126**. For brevity,

the description set forth below of the rolling portion **1400** only with reference to the lateral wall **1125**, however, this description also applies equally to the lateral wall **1126**.

(125) With reference to FIG. **19**, as noted above with reference to the seal **114** and FIG. **7a**, the medial portion **1256** comprises an upper portion **1256a** and a lower portion **1256b**. The upper portion **1256a** can be angled, with respect to plane P, at approximately 10 degrees, 0-15 degrees, 0-5 degrees, 5-10 degrees, 10-15 degrees, 5-15 degrees, greater than 0 degrees, or various sub-ranges of 0-15 degrees. The lower portion **1256b** of the medial portion **1256** can be angled, with respect to plane P, at approximately 20 degrees, 0-30 degrees, 0-20 degrees, 10-30 degrees, 15-25 degrees. In some embodiments, the upper portion **1256a** of the medial portion **1256** comprises up to one half of the medial portion **1256**.

(126) As viewed in the cross-section of FIG. **16**, the rolling portion **1400** can comprise a concave portion of the lateral wall **1125**. With reference to FIG. **12**, the concave portion **1400** can extend across the lateral wall **1125**, in the circumferential direction around the seal **1114**. Optionally, portions of the concave portion **1400** can extend onto portions of the distal wall **1122** and the proximal wall **1124**. As noted above, the lateral **1126** can include the same or similar concave portion **1400**.

(127) By including a concave portion **1400** along the lateral wall **1125**, between the nostril prong **120** and the rolling portion **306**, the concave portion **1400** can help facilitate a more dispersed and larger magnitude rolling deformations. For example, with reference to FIG. **19**, when a user is wearing a mask including the seal **1114**, a lateral force F can cause deformation of the seal **1114**. The force F can be caused by numerous different sources. For example, the force F can be generated when a user sleeps on their side with the side of their head resting on a pillow. In such a position, the pillow can push on a side of the seal **1114**, frame **116** or the conduit **106**. The force F can be caused in other ways as well.

(128) In response to the force F, the seal **1114** can deform, for example, the inlet orifice **1130** can be pushed laterally towards the right, as viewed in FIG. **19**, causing the side wall **1126** to deform, allowing the inlet orifice **1130** to move laterally relative to the nostril prongs **120**, **121**. This is also known as a “racking” motion. The concave portions **1400** can help allow the side walls **1125**, **1126** to rack in response to the force F and thus more readily allow the motion to be accommodated without collapsing the seal **1114**. In addition to “racking” motion, there may be some rolling or rotating (not shown) of the inlet orifice **1130** in response to the force F.

(129) With reference to FIGS. **14** and **15**, the seal **1114** can also include a thickened area **1410**. As shown in FIG. **14**, the thickened area **1410** can be approximately centered in the lateral walls **125**, **126**. Thus, as shown in the sectional view of FIG. **14**, as the lateral wall **126** extends from the proximal wall **1124** toward the distal wall **1122**, the thickness of the lateral wall **1126** starts from a thinner thickness adjacent to the proximal wall **1124**, thickens in the thickened portion **1410**, then reduces thickness again in the portions approaching the distal wall **1122**. In some embodiments, the portions adjacent to the proximal wall **1124** can be in the range of 0.7 to 0.9 millimeters. The thickened portion can have a thickness of 1.2 millimeters, and the portions of the lateral wall **1126** adjacent the distal wall **1122** can have a thinner thickness, for example, 0.7 millimeters. The above mentioned thicknesses are merely example and are not intended to be limiting.

(130) With reference to FIGS. **20-23**, the thickened portion **1410** of the seal **1114** can be roughly circular and can optionally be disposed within the concave portion **1400**. As shown in the various views of FIGS. **20-23**, the areas indicated by shading **1420** can be approximately 0.75 millimeters, the thickness of the region shaded **1422** can be approximately 0.9 millimeters, and the thickness of the region shaded **1424** can be approximately 1.1 millimeters. These thicknesses are merely examples of an embodiment, and other thicknesses can be used. As shown in FIGS. **21-23**, additional thickened areas can be disposed near the rolling portion **1300**.

(131) The additional rolling structure provided to the lateral walls **1125**, **1126** can provide for additional comfort and sealing performance. The concave portion **1400** and/or the thickened

portion **1410** can provide an additional rolling section that forms a cushion and can act as a shock absorber, such that movement of the sealing surfaces extending around the nostril prongs **120**, **121** and the movements of the frame **116** can be more isolated from one another. The concave portion **1400** can extend on portions of the seal **1114** forward from the patient's face and on lateral walls **1125**, **1126**, as described above.

(132) The concave sections **1400** can be configured to provide more stability under lateral, side-to-side movement of the seal **1114**, such as that illustrated and described above with reference to FIG. **19**. An additional benefit is that the concave portions **1400** can help reduce the overall visual appearance of the sides of the seal **1114**, improving patient acceptance.

(133) Seal with Bellows

(134) FIGS. **32-421** illustrate an example embodiment of a seal **500** that includes a sealing portion **510** and a lower bellows portion **520**. The sealing portion **510** includes nostril prongs **120**, **121** and sealing surfaces **512** that form seals with the nostrils of a user during use, which may be similar to other embodiments shown and described herein. The lower bellows portion **520** supports the sealing portion **510**. The lower bellows portion **520** includes a bellows structure **522** and the inlet orifice **130**.

(135) The bellows structure **522** extends circumferentially around a portion or an entirety of a base and/or the inlet orifice **130** of the seal **500**. The bellows structure **522** allows for linear compression and/or expansion of the seal **500** in an axial direction (e.g., along an axis extending through a center of the inlet orifice **130** and between the prongs **120**, **121** as shown by the dashed lines in FIG. **33**) as shown by outline **524** in FIG. **34**. Such compression and/or expansion can advantageously compensate for and/or isolate hose drag (i.e., forces applied to the seal **500** by a conduit coupled to the inlet orifice **130**) and/or can advantageously allow for adaptation to differing facial geometry of different users. In use, the lower bellows portion **520**, e.g., the bellows structure **522**, is configured to be in a compressed state when the seal **500** is fitted to the user's face. Therefore, if hose drag occurs, the lower bellows portion **520** expands away from the user's face before the sealing surfaces **512** and/or prongs **120**, **121** are pulled away from the user's face, which can cause or allow for leaks between the seal **500** and user's face. A greater range of possible compression and extension of the lower bellows portion **520** allows the seal **500** to withstand a greater degree of hose drag. In some embodiments, an initial (i.e., in a relaxed state in which the seal **500** has not been expanded, compressed, splayed, or inflated as described herein) height of the seal **500** (indicated by **h1** in FIG. **34**) is 32 mm. In some embodiments, when the seal **500** is fully or maximally compressed, a connector coupled to the inlet orifice **130** can be displaced (where the amount of displacement is indicated by **h2** in FIG. **34**) relative to the sealing surface **512** by up to 12 mm without adversely affecting the performance of the sealing surface **512**.

(136) The bellows structure **522** may be any suitable bellows arrangement that permits relative movement between the base and the prongs **120**, **121**, such as the movement described immediately above or elsewhere herein. For example, the bellows structure **522** may be a structure of reduced cross sectional area or, alternatively, may be a thinner section on the inner side of the wall, or a combination thereof. The bellows section may have a planar outer wall shape/structure (or a curved shape that is continuous with portions of the seal **500** outside of the bellows section) and may have an internal thinning of the wall to result in the same/similar function as the bellows structure **522**.

(137) A further alternative arrangement is that the bellows section could be a convex bellows rather than a concave section as shown. The current illustrated embodiment shows a bellows section **522** or a pleat that is concave in shape, but alternatively the bellows section **522** may be a pleat that is convex and extends outwardly. In at least some configurations, the convex bellows would expand outwardly when compressed. Such a convex bellows section can act as a support to prevent the seal from over rotating about the center of the bellows.

(138) The sealing portion **210** can pivot around the center of the bellows structure **522** (for example, as indicated by the curved arrows in FIG. **33**) such that the sealing portion **210** can be

angled relative to the lower bellows portion **520**. Such pivoting can allow the seal **500** to compensate for angled forces applied to the seal **500** by the conduit attached to the inlet orifice **130** and can help inhibit or reduce disturbance to the sealing surface **512** that may result from such angled forces. The pivoting can also allow the seal **500** to adjust for and accommodate variability in the angle of the user's lower nasal surfaces for different users.

(139) In some embodiments, the center of the inlet orifice **130** and center of the sealing surface **512** are aligned or approximately aligned, as indicated by the horizontal arrows and dashed lines in FIG. **33**. Such alignment can advantageously encourage the bellows structure **522** to compress rather than form a hinge about which the sealing portion **510** and lower bellows portion **520** bend relative to each other. The forces applied to the inlet orifice **130** and the sealing surface **512** can be offset, but oriented in the same direction or parallel to one another. In some configurations, the offset is small relative to a size of the seal **500**, such as relative to a cross-sectional dimension of the connector opening **130** or interior cavity of the seal **500**.

(140) As shown, the seal **500** can have a rounded cross-sectional profile. In other words, the seal **500** may not have sharp corners. A rounded cross-sectional profile can advantageously promote a smooth compression motion of the seal **500** and a substantially linear increase in force as displacement or compression of the seal **500** increases during at least a portion of the range of compression, as shown in the force vs. extension graph of FIG. **35**.

(141) In the illustrated embodiment, the seal **500** has a relatively large inlet orifice **130** in comparison with an overall size of the seal **500** (e.g., cross-sectional dimension of the cavity or other portion of the seal **500**). A relatively large inlet orifice **130** can allow the seal **500** to be removed from a mold tool more easily during manufacturing. As shown, a perimeter of the inlet orifice **130** is inwardly offset from an outer perimeter of the lower bellows portion **520**. This offset is large enough such that the inlet orifice **130** does not significantly interfere with the function of the lower bellows portion **520** and bellows structure **522**.

(142) As shown by the generally horizontal arrow in FIG. **36**, lateral sides of the lower bellows portion **520** can curve toward the sealing portion **510** (upward in the orientation of FIG. **36**). In the illustrated embodiment, a curvature of the lower bellows portion **520** matches or corresponds to (or substantially or generally matches or corresponds to) a curvature of the lower surfaces of the sealing portion **510**. In such embodiments, a distance between the lower bellows portion **520** and the lower surfaces of the sealing portion **510** (indicated by the generally vertical arrows in FIG. **36**) is substantially the same or equal around an entire circumference of the seal **500**. In some embodiments, the distance between the lower bellows portion **520** and the lower surfaces of the sealing portion **510** is 6 mm. A constant distance between the lower bellows portion **520** and lower surfaces of the sealing portion **510** around the circumference of the seal **500** can advantageously allow for a more even compression of the seal **500** around the entire lower bellows portion **520**. In other words, the seal **500** is less likely to bottom out in one area of the seal **500** before a remainder of the circumference of the seal **500**. The curvature of the lower bellows portion **520** can also give the seal **500** a smaller appearance.

(143) The perimeter of the lower bellows portion **520** and bellows structure **522** can be smaller than the perimeter of the sealing portion **510** as shown in FIGS. **37-38**. A relatively smaller lower bellows portion **520** can advantageously help reduce or minimize the size of the seal **500**. As shown in FIG. **37**, the lower bellows portion **520** can offset (indicated by arrow **523**) from a back of the sealing portion **510** (e.g., from a lowest, in the orientation of FIG. **37**, point of proximal wall **124**). Such an offset can inhibit the lower bellows portion **520** from contacting the user's lip in use. In some embodiments, the sealing portion **510** has a greatest width (indicated by W4 in FIG. **38**) of 43 mm, the inlet orifice **130** has a width (indicated by W1 in FIG. **38**) of 28 mm, the lower bellows portion **520** has a greatest width (indicated by W2 in FIG. **38**) of 34 mm, and the bellows structure **522** has a width (indicated by W3 in FIG. **38**) of 26.5 mm. Such dimensions are exemplary and may vary. However, in some configurations, the relative sizes can be as illustrated.

(144) In some embodiments, the sealing portion **510** and/or prongs **120**, **121** can flex outwardly or away from each other as indicated by dashed lines showing a flexed or splayed prong profile in FIG. **39**. Flexing of the prongs **120**, **121** or sealing portion **510** can advantageously allow the seal **500** to accommodate users having relatively large nares and/or nares that are spaced apart by a relatively wide septum. Reducing or minimizing a height of a portion of the sealing portion **510** not including the prongs **120**, **121** (indicated by “h” in FIG. **39**) advantageously allows for a tight, continuously curved outer surface in the sealing portion **520**, which can increase the ability of the prongs **120**, **121** to splay outwards without the outer surfaces of the seal **500** collapsing as a shorter, more curved surface is generally less likely to buckle and/or collapse. In some embodiments, the height h is 19 mm. In some embodiments, the prongs **120**, **121** can splay outwardly away from each other by 2 mm or more. In some embodiments, a distance between innermost edges of the prongs **120**, **121** in a relaxed state is 7.5 mm and a distance between the innermost edges of the prongs **120**, **121** in a splayed state is 9.5 mm.

(145) In some embodiments, the seal **500** is semi-inflatable. When the prongs **120**, **121** are engaged and sealed with the user's nares and CPAP pressure is applied in use, the seal **500** inflates slightly. The seal **500** can extend or increase in height (along a direction parallel to an axis extending through the center of the inlet orifice **130**, indicated by the dashed lines in FIG. **33**) and/or the prongs **120**, **121** can splay apart from each other as shown by the inflated profile outline shown in FIG. **40**. Such inflation can advantageously allow the prongs **120**, **121** to engage the user's nares more securely and improve the seal with the user's nares. In some embodiments, the seal **500** can increase in height by 2 mm or more when inflated. In some embodiments, the prongs **120**, **121** can splay apart from each other by 2 mm or more when the seal **500** is inflated.

(146) In some embodiments, a thickness of the bellows structure **522** is consistent or constant throughout the bellows structure **522**. The thickness of the bellows structure **522** can be 0.75 mm. In some embodiments a thickness of the lower bellows portion **520** is consistent or constant throughout the lower bellows portion **520**. A constant thickness can advantageously provide unbiased compression over the entirety of the bellows structure **522** and/or lower bellows portion **520**. In such an embodiment, the cross-sectional profile of the seal **500** would therefore have a more significant impact on how compression of the seal **500** occurs than the thickness of the seal **500**.

(147) In some embodiments, the seal **500** includes a thickened region **530** as shown in FIG. **41** that has a greater cross-sectional thickness than a remainder of the seal **500**. In the illustrated embodiment, the thickened region **530** extends along lateral front portions of the seal **500**. The thickened region **530** can help prevent or inhibit the sealing surface **512** from collapsing in the thickened region **530**. In some embodiments, the thickened region **530** has a thickness of 1 mm and a remainder of the seal **500** has a thickness of 0.75 mm or similar relative thicknesses.

(148) Alternatively, the bellows section may have a varying thickness. The base section of the bellows may be thickest and the portion of the bellows at the top, i.e., closest to the prong or pillow is thinnest, or vice-versa. The thin portion of the bellow allows for greater flexing/pivoting. The thicker base allows for greater stability at the base.

(149) In a further alternative, the bellow may be thick at the base and at the top and thin in the middle. The middle being thinner than the top and the bottom of the bellow. This allows for greater flexibility and flexure in the middle section of the bellow, while the thicker section provide for greater support at the top and bottom of the seal, allowing the prong to maintain its shape and also maintain orientation.

(150) In some embodiments, an inner volume of the lower bellows portion **520**, indicated by area 3 in FIG. **421**, is about 3966 mm³. In some embodiments, an inner volume of sealing portion **510** is about 11,158 mm³. An inner volume of each half of the sealing portion **510** (indicated by areas 1 and 2 in FIG. **421** where the sealing portion **510** is separated by a plane extending between the prongs **120**, **121**) can be 5579 mm³. Thus, the areas 1 and 2 can be collectively greater than the

volume 3. In some configurations, each of the volumes 1 and 2 can be greater than the volume 3.

(151) Seal with Rolling Section

(152) FIGS. 43-54C illustrate an example embodiment of a seal **600** that includes a rolling section **620** between the sealing surface **610** and a connector receiving portion **630**. The rolling section **620** can advantageously at least partially isolate the sealing surface **610** from hose drag forces. The rolling section **620** and sealing surface **610** are delimited or separated by a stiffener or stiffening member, such as a rib **640**, that projects inwardly from an inner surface of the seal **600**. The rolling section **620** is configured to roll over the connector receiving portion **630**, as indicated by the dashed line profile in FIG. 44, to absorb or accommodate for forces on the connector receiving portion **630** and inhibit deformation of the sealing surface **610**. In some embodiments, a height of the seal **600** in a neutral state, indicated by h1 in FIG. 44, is 34.5 mm. In some embodiments, a height of the seal **600** can decrease by approximately 5 mm or more when the rolling section **620** is rolled over the connector receiving portion **630**. In some embodiments, the height of the seal **600** in a compressed state in which the rolling section **620** is rolled over the connector receiving portion **630**, indicated by h2 in FIG. 44, is approximately 29.9 mm. In other configurations, the height may differ from that described above, but the relative amount of roll can remain the same.

(153) The rib **640** can advantageously separate the rolling section **620** from the sealing surface **610** to help isolate movement of the connector receiving portion **630** from the sealing surface **610** and reduce or minimize the effect of hose drag. The rib **640** and/or rolling section **620** can also advantageously allow the sealing surface **610** to be positioned at differing angles relative to the connector receiving portion **630** such that the seal **600** can adapt to and accommodate varying facial geometries of different users. In some embodiments, the rib **640** is positioned approximately halfway between the sealing surface **610** (e.g., a portion of the sealing surface **610** between the prongs **120**, **121**) and the connector receiving portion **630**.

(154) The rib **640** can extend around an entire inner perimeter of the seal **600**. In other embodiments, the rib **640** includes a gap **642** extending along at least a portion of the front of the seal **600** as shown in FIGS. 43-44. Once the rolling section **620** rolls over the connector receiving portion **630** to a certain extent, the rib **640** contacts internal surfaces of the rolling section **620** and/or the connector receiving portion **630** to prevent or inhibit further movement. In embodiments in which the rib **640** includes a gap **642** along the front of the seal **600**, the gap allows the rolling section **620** in the front of the seal **600** to roll farther over the connector receiving portion **630** than the back of the seal **600**. This allows for a greater travel or range of motion of the rolling portion **620** in the front of the seal **600** and the rolling section **620** in the front of the seal **600** may therefore not bottom out.

(155) In some embodiments, the seal **600** is configured to be fitted to the user in use such that the rolling section **620** is at least partially rolled over the connector receiving portion **630** at least a majority of the time. This advantageously allows the connector receiving portion **630** to be pulled away from the sealing surface **610** to some extent without the sealing surface **610** losing contact with the user's nares.

(156) In some embodiments, the rib **640** is made of silicone. In some embodiments, the rib **640** has an inner thickness (indicated by dimension Y in FIG. 45) in the range of 1 mm-4 mm, for example, 1.7 mm, an outer thickness (indicated by dimension Z in FIG. 45) in the range of 1 mm-5 mm, for example, 2.5 mm, and a length (indicated by dimension X in FIG. 45) in the range of 1 mm-6 mm, for example, 3 mm. The inner thickness can be less than the outer thickness. The length can be greater than the inner and outer thicknesses. In some embodiments, the rib **640** has rounded edges.

(157) A thickness of the rolling section **620** can gradually transition between the rib **640** and the connector receiving portion **630** as shown in FIG. 46. For example, in some embodiments, the thickness transitions from 0.5 mm proximate or adjacent the connector receiving portion **630** to 0.8 mm proximate or adjacent the rib **640**. In other configurations, the actual dimension may differ, but the proportions can be the same. The transitioning thickness advantageously allows the rolling

section **620** to have a spring force that increases as the seal **600** becomes more compressed, which can help prevent or inhibit sharp increases in force that may be caused by the seal **600** suddenly bottoming out. This can advantageously improve user comfort. The force on the seal **600** increases substantially linearly as the rolling section **620** is displaced and remains relatively low until the rolling section **620** bottoms out, for example as shown in the theoretical force vs. displacement profile shown in FIG. **47** and force profile shown in FIG. **48**. The thickness of the seal **600** can be consistent or constant throughout the sealing surface **610**. In some embodiments, the thickness of the sealing surface **610** is 0.75 mm. FIG. **49** shows a heat map showing the generally constant thickness in the sealing surface **610** and varying thickness at the rib **640**, rolling section **620**, and connector receiving portion **630**.

(158) In some embodiments, the connector receiving portion **630** is generally kidney bean shaped and follows the overall shape of the seal **600** as shown in FIGS. **50A-50C**. In such embodiments, the rolling section **620** extends outwardly from the connector receiving portion **630** a substantially or relatively more uniform depth or distance (indicated by the arrows in FIG. **50A** as well as D1 and D2 in FIG. **51**) around the entire perimeter or circumference of the seal **600**. In contrast, if the connector receiving portion **630** had a generally oval shape, indicated by the dashed line in FIG. **50B**, the rolling section **620** would have an uneven depth or distance as indicated by the arrows in FIG. **50B**. A relatively more uniform depth of the rolling section **620**, as provided by the kidney bean shaped connector receiving portion **630**, advantageously allows the seal **600**, e.g., the rolling section **620**, to roll more evenly across the entire lower half of the seal **600** or around an entire perimeter or circumference of the seal **600**.

(159) In some embodiments, the prongs **120**, **121** can splay apart from each other to accommodate varying septum widths of different users as shown in FIG. **52**. The rolling section **620** advantageously allows the prongs **120**, **121** to splay apart without the sealing surfaces **610** collapsing or collapsing to a significant extent.

(160) The connector receiving portion **630** extends from an intersection **634** with the rolling section **620** to an open base end **632** configured to receive, for example, a conduit or connector. In the illustrated embodiment, the open base end **632** and intersection **634** with the rolling section **620** are curved. The curvature of the intersection **634** can encourage consistent movement of the rolling section **620** around the perimeter or circumference of the seal **600**. The curvature of the intersection **634** and/or base end **632** can also make the seal **600** shorter and/or give the seal **600** a relatively smaller appearance compared to a seal having a flat or straight intersection and/or base end. In some embodiments, the curvature is such that a distance (indicated by h1 in FIG. **53A**) between a center of the base end **632** or intersection **634** and a lateral side of the base end **632** or intersection **634**, respectively, along a line parallel to an axis extending through the center of the connector receiving portion **630** and between the prongs **120**, **121** is approximately 3.26 mm.

(161) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising”, and the like, are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, that is to say, in the sense of “including, but not limited to”.

(162) Reference to any prior art in this specification is not, and should not be taken as, an acknowledgement or any form of suggestion that that prior art forms part of the common general knowledge in the field of endeavor in any country in the world.

(163) The inventions disclosed herein may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, in any or all combinations of two or more of said parts, elements or features.

(164) Where, in the foregoing description reference has been made to integers or components having known equivalents thereof, those integers are herein incorporated as if individually set forth.

(165) It should be noted that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and

modifications may be made without departing from the spirit and scope of the inventions and without diminishing its attendant advantages. For instance, various components may be repositioned as desired. It is therefore intended that such changes and modifications be included within the scope of the inventions. Moreover, not all of the features, aspects and advantages are necessarily required to practice the present inventions. Accordingly, the scope of some of the present inventions is intended to be defined only by the claims that follow.

Claims

1. A nasal seal comprising: a seal body formed of a soft flexible material and defining an inner cavity that receives a supply of breathing gases, the seal body comprising: a first nostril prong; a second nostril prong; wherein each of the first nostril prong and the second nostril prong are non-conical and asymmetrical; and a rolling portion defined by a sidewall portion of the seal body and configured to roll over on itself, the rolling portion comprising an upper portion configured to be distal of an upper lip of a user and a lower portion configured to be proximate the upper lip of the user, wherein the upper portion provides less resistance to roll than the lower portion.
 2. The nasal seal of claim 1, wherein the seal body defines a supply opening through which the supply of breathing gases passes into the inner cavity, wherein the supply opening is non-circular in shape.
 3. The nasal seal of claim 2, wherein the supply opening is bean-shaped.
 4. The nasal seal of claim 1, wherein each of the first nostril prong and the second nostril prong comprises a medial portion, wherein the medial portion is oriented at an angle relative to a central plane of the seal body.
 5. The nasal seal of claim 4, wherein the medial portion comprises an upper portion and a lower portion, wherein the angle relative to the central plane of the lower portion is greater than the angle relative to the central plane of the upper portion.
 6. The nasal seal of claim 1, wherein a proximal wall of the seal body comprises a concave portion extending in a lateral direction of the seal body and having a shape that generally conforms to a shape of an upper lip of a user.
 7. The nasal seal of claim 1, wherein the first nostril prong comprises a first sealing surface and the second nostril prong comprises a second sealing surface, wherein portions of the first sealing surface and the second sealing surface are convex.
 8. The nasal seal of claim 1, wherein the seal body comprises a central wall portion extending between and connecting the first nostril prong and the second nostril prong, wherein the central wall portion is spaced deeply from an end surface of the first nostril prong and second nostril prong such that contact between the central wall portion and a septum of a user is reduced or eliminated.
 9. The nasal seal of claim 1, wherein the upper rolling portion has a radial dimension that is larger than a corresponding radial dimension of the lower rolling portion.
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